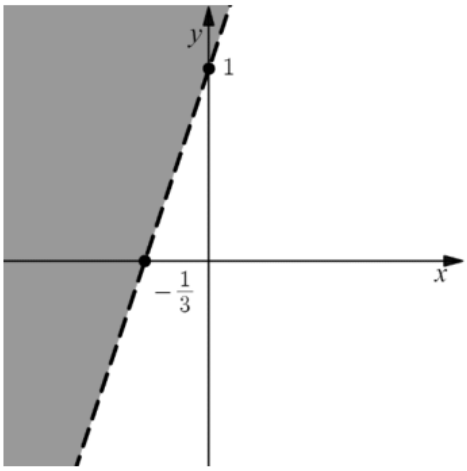
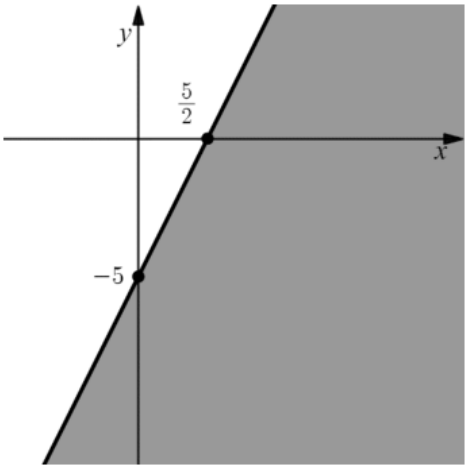
1.21 Inequalities: Identifying Regions

Worked Solutions

1 a)

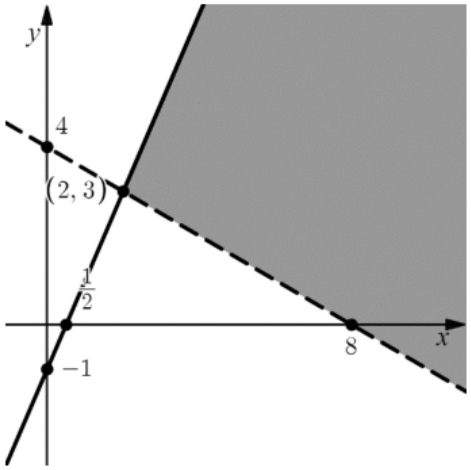


1 b)



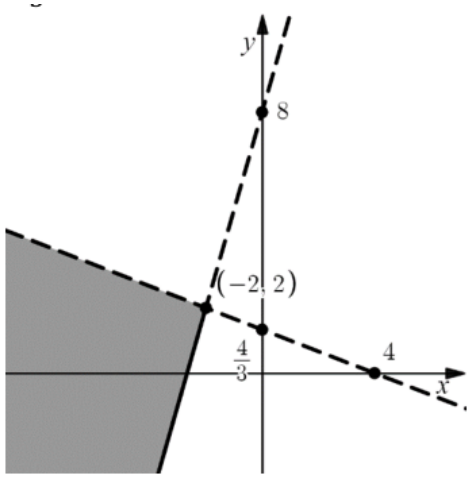
2)

$x + 2y = 8 \Rightarrow y = -\frac{1}{2}x + 4$
$2x - 1 = -\frac{1}{2}x + 4 \Rightarrow \frac{5}{2}x = 5 \Rightarrow x = 2, y = 3$

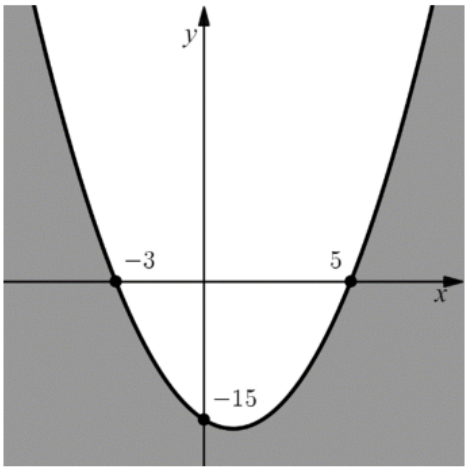


3)

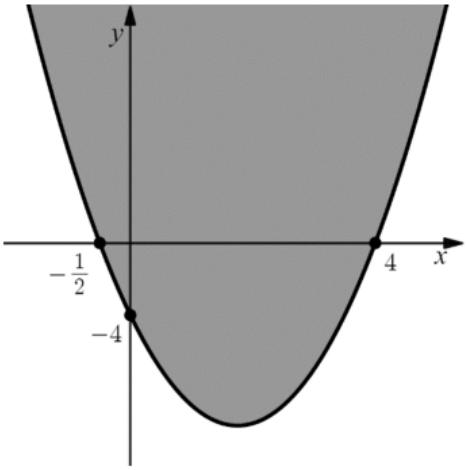
$x + 3y = 4 \Rightarrow y = -\frac{1}{3}x + \frac{4}{3}$
$-\frac{1}{3}x + \frac{4}{3} = 3x + 8 \Rightarrow \frac{10}{3}x = -\frac{20}{3} \Rightarrow x = -2, y = 2$



4 a)

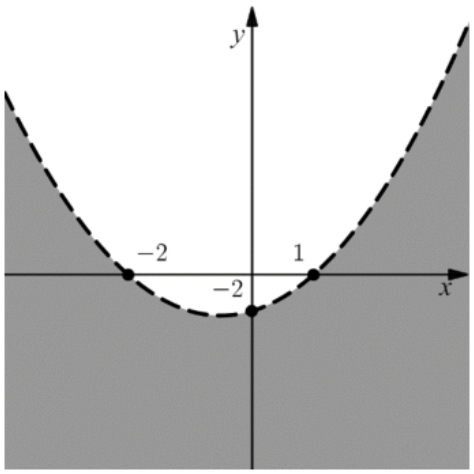


4 b)



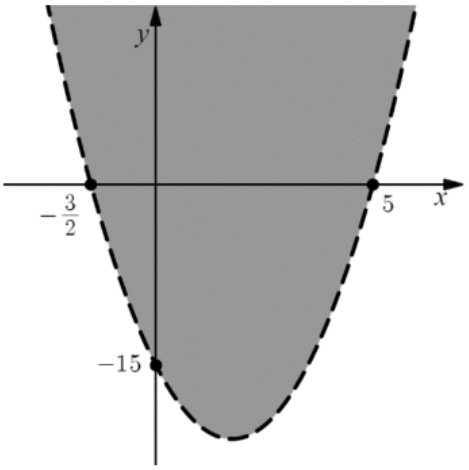
4 c)

$y < (x+2)(x-1)$



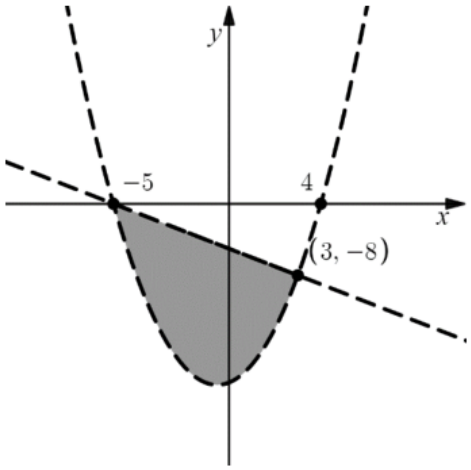
4 d)

$y > (2x+3)(x-5)$



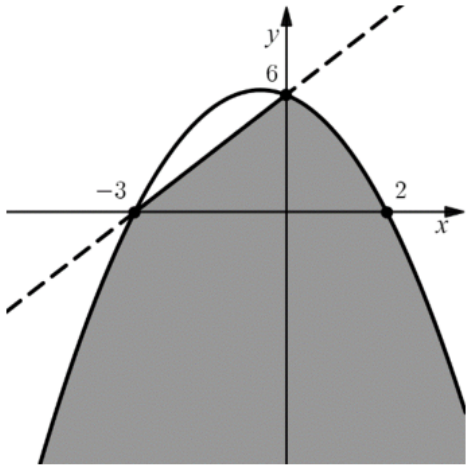
5 a)

$x^2 + x - 20 = (x + 5)(x - 4)$
$y + x = -5 \Rightarrow y = -x - 5$
$x^2 + x - 20 = -x - 5$
$\Rightarrow x^2 + 2x - 15 = (x + 5)(x - 3)$
$\Rightarrow (-5, 0)$ and $(3, -8)$



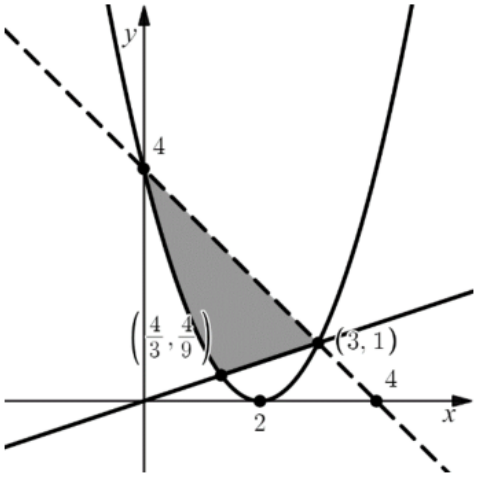
5 b)

$-x^2 - x + 6 = -(x + 3)(x - 2)$
$y = 2x + 6$
$-x^2 - x + 6 = 2x + 6 \Rightarrow 0 = x^2 + 3x$
$\Rightarrow (0, 6), (-3, 0)$



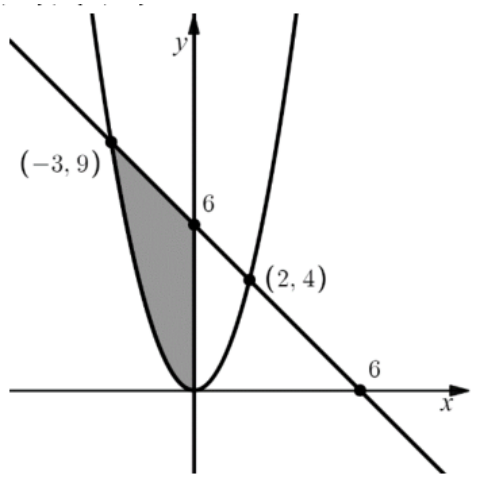
6)

$y = \frac{1}{3}x \Rightarrow \frac{1}{3}x = (x-2)^2$
$\frac{1}{3}x = x^2 - 4x + 4 \Rightarrow 3x^2 - 13x + 12 = 0$
$\Rightarrow (x-3)(4x-3) = 0 \Rightarrow (3,1), (\frac{4}{3}, \frac{4}{9})$



7)

$y = 6 - x \Rightarrow 6 - x = x^2 \Rightarrow x^2 + x - 6 = 0$
$\Rightarrow (x-2)(x+3) = 0 \Rightarrow (-3,9), (2,4)$



8 a)

$y > x + 1$ and $y \leq 6 + 5x - x^2$.

8 b)

$6 + 5x - x^2 = x + 1 \Rightarrow x^2 - 4x - 5 = 0$
$\Rightarrow (x-5)(x+1) = 0$
$x = 5, x = -1$
P is $(-1, 0)$ and Q is $(5, 6)$.

8 c)

$-1 < x < 5$
$\{x : x \in \mathbb{R}, -1 < x < 5\}$.

9 a)

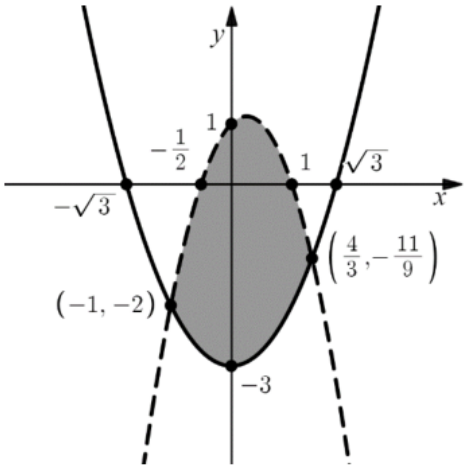
$y = (x-3)(x-4) = x^2 - 7x + 12$
$y = -(x+2)(x-4) = -x^2 + 2x + 8$
So the inequalities are:
$y \geq x^2 - 7x + 12$ and $y < -x^2 + 2x + 8$.

9 b)

$x^2 - 7x + 12 = -x^2 + 2x + 8$
$\Rightarrow 2x^2 - 9x + 4 = 0 \Rightarrow (x-4)(2x-1) = 0$
$x = \frac{1}{2}, y = \frac{35}{4}$
P is $(\frac{1}{2}, \frac{35}{4})$.

10 a)

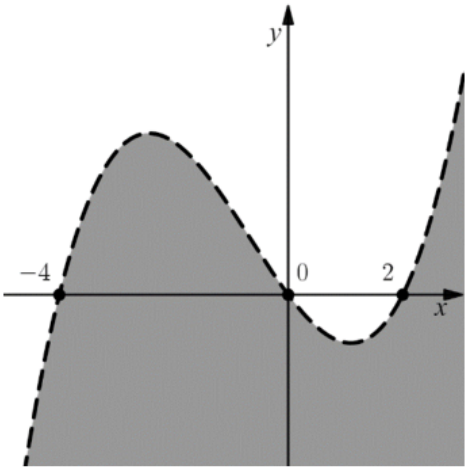
$y = x^2 - 3$ and $y = -2x^2 + x + 1$
$x^2 - 3 = -2x^2 + x + 1 \Rightarrow 3x^2 - x - 4 = 0$
$\Rightarrow (3x-4)(x+1) = 0 \Rightarrow (-1, -2), (\frac{4}{3}, -\frac{11}{9})$



10 b)

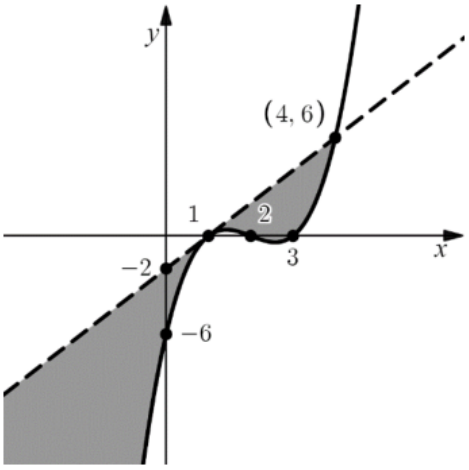
$-1 < x < \frac{4}{3}$
$\{x : x \in \mathbb{R}, -1 < x < \frac{4}{3}\}$.

11)



12)

$x^3 - 6x^2 + 11x - 6 = (x-1)(x-2)(x-3)$
$x^3 - 6x^2 + 11x - 6 = 2x - 2$
$\Rightarrow x^3 - 6x^2 + 9x - 4 = 0$
$\Rightarrow (x-4)(x-1)^2 = 0 \Rightarrow (1,0)$ and $(4,6)$

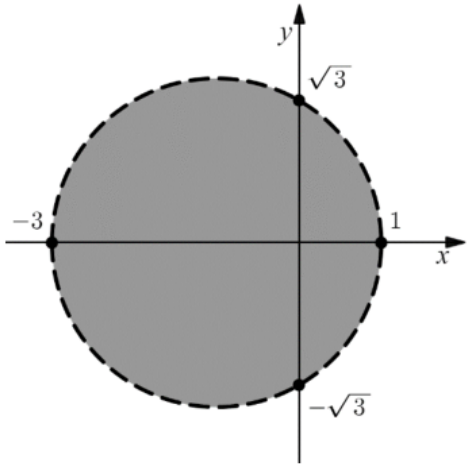


13 a)

$(x+1)^2 - 1 + y^2 = 3$
$(x+1)^2 + y^2 = 4$ which is a circle with centre $(-1, 0)$ and radius 2.

13 b)

When $x = 0, y = \pm\sqrt{3} \Rightarrow (0, -\sqrt{3}), (0, \sqrt{3})$
When $y = 0, x^2 + 2x - 3 = (x - 1)(x + 3)$
$\Rightarrow (1, 0), (-3, 0)$



14)

$(x-3)^2+(y-1)^2=25$ has centre $(3,1)$
When $x=0, (-3)^2+(y-1)^2=25 \Rightarrow y=1\pm 4$
$\Rightarrow (0,-3)$ and $(0,5)$
When $y=0, (x-3)^2+(-1)^2=25 \Rightarrow y=3\pm 2\sqrt{6}$
$y=3-x \Rightarrow (x-3)^2+(3-x-1)^2=25$
$\Rightarrow x^2-6x+9+4-4x+x^2=25$
$\Rightarrow 2x^2-10x-12=0 \Rightarrow 2(x-6)(x+1)=0$
$\Rightarrow (-1,4)$ and $(6,-3)$

